

Sequencing Data in the Cloud

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Section 1. Proportion of G/G

Download a CSV file from ENSEMBLE. < https://uswest.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39815097-39975098;v=1s8067378;vdb=variation;vf=88214830#373531_tablePanel
>

```
mxl<- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

	Sample..Male.Female.Unknown.	Genotype..forward.strand.	Population.s.	Father
1	NA19648 (F)	A A	ALL, AMR, MXL	-
2	NA19649 (M)	G G	ALL, AMR, MXL	-
3	NA19651 (F)	A A	ALL, AMR, MXL	-
4	NA19652 (M)	G G	ALL, AMR, MXL	-
5	NA19654 (F)	G G	ALL, AMR, MXL	-
6	NA19655 (M)	A G	ALL, AMR, MXL	-
Mother				
1	-			
2	-			
3	-			
4	-			
5	-			
6	-			

```
table(mxl$Genotype..forward.strand.)
```

```
A|A A|G G|A G|G
22  21  12   9
```

```
table(mx1$Genotype..forward.strand.)/nrow(mx1)*100
```

A A	A G	G A	G G
34.3750	32.8125	18.7500	14.0625

Now we will look at another population GBR.

```
gbr<-read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

Now find proportion of G|G

```
round(table(gbr$Genotype..forward.strand.)/nrow(gbr)*100,2)
```

A A	A G	G A	G G
25.27	18.68	26.37	29.67

This variant associated with childhood asthma is more frequent in GBR population.